

Title

Adding synchronous therapeutic contact to a smoking-cessation app: results of a randomized effectiveness trial comparing app-only versus app-plus-cognitive-behavioral-support models

Abstract

Remotely-delivered smoking cessation interventions expand access to evidence-based treatment, but the incremental value of human therapeutic engagement over autonomous app-based self-management remains uncertain. A randomized three-arm trial enrolled 420 adults (mean age ~48 years) actively seeking smoking abstinence support. Participants were randomly assigned to eight weeks of: (1) app-only self-guided use (n=140); (2) app combined with eight weekly group cognitive-behavioral therapy sessions via secure video (n=140); or (3) app combined with four biweekly group sessions (n=140). Primary outcome was carbon-monoxide-validated 7-day point-prevalence smoking abstinence at week eight. The app-only group achieved 18% validated abstinence at treatment completion. The high-intensity contact arm (eight weekly sessions) achieved 36% validated abstinence, while the low-intensity arm (four biweekly sessions) achieved 28% abstinence. Both therapeutic-contact arms significantly outperformed the app-only condition. At 12-month follow-up, the high-intensity group maintained substantially higher rates of sustained abstinence compared with other study arms. These findings demonstrate that supplementing digital smoking-cessation tools with regular human therapeutic guidance substantially improves both short-term and long-term abstinence rates.